Lesson 2.3 — Factored Form and Zeros

Big idea: Factored form $y = a(x - r_1)(x - r_2)$ shows you the zeros for free. The vertex sits exactly between them.

Quick review

Factored form: $y = a(x - r_1)(x - r_2)$. The zeros are r_1 and r_2.
Axis of symmetry: midpoint of the zeros: $x = (r_1 + r_2)/2$.
From factored → standard: FOIL.
From standard → factored: factor or use the quadratic formula to get the zeros, then write $a(x - r_1)(x - r_2)$.

Worked example

Worked example. $y = (x - 2)(x + 6)$. Find zeros, axis of symmetry, vertex, y-intercept, and standard form.

Zeros: $x = 2$, $x = -6$. Axis: $x = (2 + (-6))/2 = -2$. Vertex: y at $x = -2$ is $(-4)(4) = -16$, so $(-2, -16)$. y-intercept: $(-2)(6) = -12$. Standard: $x^2 + 4x - 12$.

Warm-ups

1. Find the zeros of $y = (x - 3)(x + 5)$.
2. Find the y-intercept of $y = (x + 1)(x - 4)$.
3. Find the axis of symmetry of $y = (x + 2)(x + 8)$.

Practice

1. Write a quadratic in factored form with zeros 4 and –7 and $a = 2$.
2. Convert $y = 2(x - 1)(x + 5)$ to standard form.
3. Find the vertex of $y = (x - 3)(x - 9)$.

Challenge

1. A parabola has zeros 2 and 8 and passes through (0, 8). Write its equation in factored form.
2. Show algebraically that for $y = a(x - r_1)(x - r_2)$, the axis of symmetry is exactly $x = (r_1 + r_2)/2$.

Answer key — Lesson 2.3

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $x = 3$, $x = -5$.
2. $(0 + 1)(0 - 4) = -4$.
3. $x = ((-2) + (-8))/2 = -5$.

Practice

1. $y = 2(x - 4)(x + 7)$.
2. $y = 2(x^2 + 4x - 5) = 2x^2 + 8x - 10$.
3. Axis: $x = 6$. Vertex: $(6, (3)(-3)) = (6, -9)$.

Challenge

1. $y = a(x - 2)(x - 8)$. Plug $(0, 8)$: $8 = a(-2)(-8) = 16a \Rightarrow a = 1/2$. So $y = (1/2)(x - 2)(x - 8)$.
2. Expand: $a[x^2 - (r_1+r_2)x + r_1r_2]$. Standard form coefficients: $A = a$, $B = -a(r_1+r_2)$. Axis: $-B/(2A) = a(r_1+r_2)/(2a) = (r_1+r_2)/2$. ✓